

Analog IC Design Implementation and Verification

Amplifiers – Week 2 Labs

**Course Version IC 23.1 ISR11.29 with SPECTRE 24.1 ISR1
Lab Manual**

Revision 2.0

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Module 1: Common Source Amplifier

Lab 1-1 Design of Common Source Amplifier

Objective: To use the basic schematic capture commands to build a Common Source Amplifier.

In this lab, you open the schematic capture environment window and add components, wires, and pins. The schematic is then checked and saved.

Creating a Library and Cellview

1. Open the Terminal window, open the directory of **basic_analog_labs**, by entering this command in the terminal window:

```
cd ~/basic_analog_labs
```

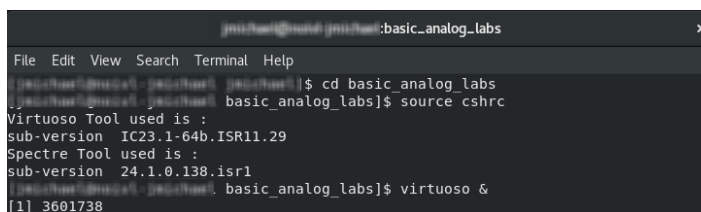
2. In that directory, source the License file by entering this command in the terminal window:

```
source cshrc
```

3. Type the following command to open CIW:

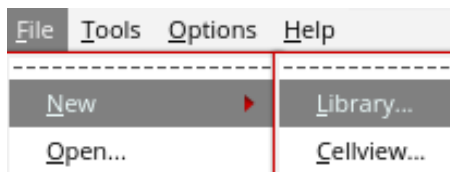
```
virtuoso &
```

The following screenshot gives the view of the terminal window on these commands executed:

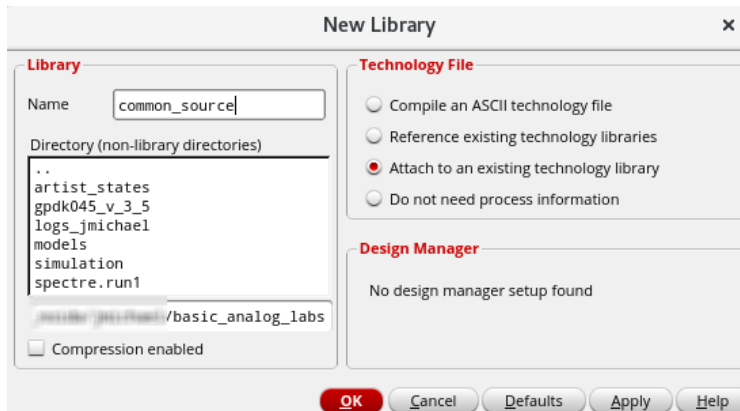


```
jmi@jmi:~/basic_analog_labs$ cd basic_analog_labs
jmi@jmi:~/basic_analog_labs$ source cshrc
Virtuoso Tool used is :
sub-version IC23.1-64b.ISR11.29
Spectre Tool used is :
sub-version 24.1.0.138.isr1
jmi@jmi:~/basic_analog_labs$ virtuoso &
[1] 3601738
```

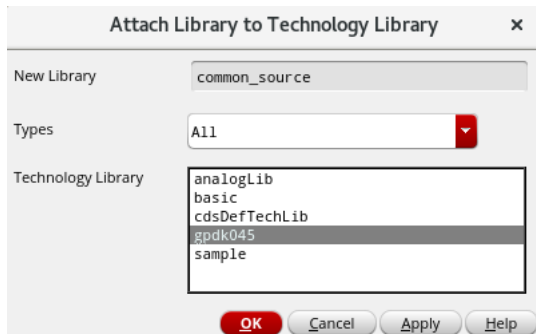
4. From CIW, choose **File – New – Library**.



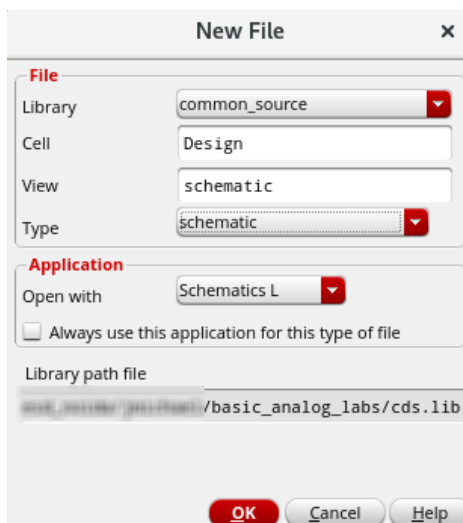
5. Name the Library as **common_source** and select the radio button for **Attach to an existing technology Library** and click on **OK**.



6. In the Attach Library to Technology Library form, select on the **gpdK045** library and click on **OK**.



7. Create a new cell view under the same **mirror** library by choosing **File – New – Cellview** created, name it as **Design**, and click on **OK** to open the schematic view.



Invoking the Add Instance Form in the Cellview

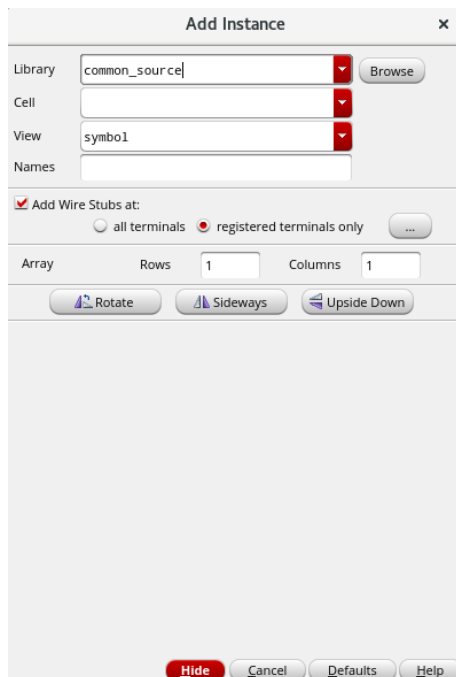
The instructions in this section describe how to build the current mirror circuit schematic shown in the following snapshot.

1. With your cursor in the schematic window, go to the top of the schematic window to the toolbar. Click the **Create Instance** icon to display the Add Instance form. 



Note: You can also choose **Create – Instance** from the pull-down menus or use the **I** key to pop up the form.

2. In the Add Instance form, click on Browse to open the Library Browser.



3. In the **Library Browser – Add Instance** form, select **gpdK045** in the library column. The Cell column of this form now displays the available cells in that library.

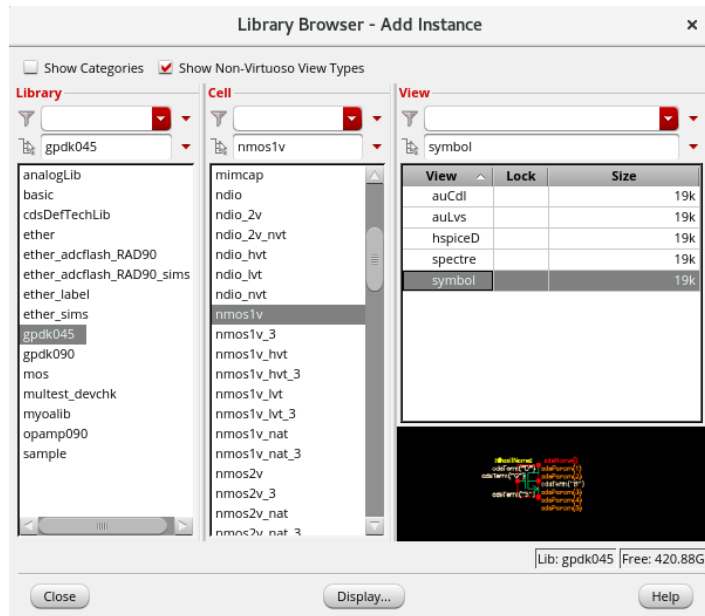
- a. Use the scroll bar to the right of the Cell column to locate **nmos1v** and **click** to select this cell. The Library Browser form now displays several views for the *nmos1v* cell.

Note: You can use the filtering option under the Cell category.

Note: You can also begin typing *nmos1v* in the *cell* name field of the Add Instance form. The auto-complete feature brings the *nmos1v* cell name into the field for easy selection.

As you make your selection, note that the Add Instance form modifies itself.

- b. Select the **symbol view** with the left mouse button. Your Library Manager should be like what you see in the following snapshot.



The Add Instance form now displays some information along with auto-filled text fields, as shown in the following snapshot. Set the parameters as follows:

Total Width: 1u

4. Close the **Library Browser – Add Instance** form. Leave the Add Instance form open with *Library*, *Cell* and *View* pull-down cyclic fields populated.

Adding Components to a Schematic

1. Click the **Hide** button toward the bottom of the form.

Follow the next steps diligently, as they will allow you to place the chosen *nmos* transistor on your schematic.

- a. Move your *cursor* into the schematic window. A highlighted symbol is displayed. It is ready to be placed on the schematic.
- b. **Left**-click to place the symbol on the schematic.
- c. Place the instance of *nmos1v* on the schematic.

Note: Observe that the device has been added to the schematic and the symbol has an instance name. A highlighted symbol of *nmos1v* remains and moves with the cursor.

- d. Click the <**Esc**> key to quit placement mode.

2. Press the **I** key.

The Add Instance form is displayed again, with the library, cell, and view fields filled. Likewise, a highlighted symbol tracks the cursor in the schematic window.

- a. Move the cursor into the Add Instance form. Change the library field to **analogLib** from the library column. Click in the cell text field and choose **vsin**.

Note: You can use the **Browse** button to search for *vsin*. However, entering *vsin* directly into the text field saves time.

- b. Set the parameter as follows:

DC voltage: 700.0m V

AC magnitude: 2m V

Frequency: 1G Hz

- c. Your form should be similar to what you see in the following snapshot:

Add Instance

Library: analogLib
Cell: vsin
View: symbol
Names:

☒ Add Wire Stubs at:
☐ all terminals ☒ registered terminals only ...

Array: Rows: 1 Columns: 1
 Rotate Sideways Upside Down

First frequency name
 Second frequency name
 Noise file name
 Number of noise/freq pairs: 0
 DC voltage: 700.0m V
 AC magnitude: 2m V
 AC phase
 XF magnitude
 PAC magnitude
 PAC phase
 Delay time
 Offset voltage
 Amplitude
 Initial phase for Sinusoid
 Frequency: 1G Hz
 Amplitude 2
 Initial phase for Sinusoid 2
 Frequency 2

Hide Cancel Defaults Help

- d. **Hide** the form. Place this instance of *vsin* on the schematic beside the instance.

3. Press the **I** key.

Under the **analogLib** library, click in the cell text field and choose **vdc** in the Add Instance form change the parameter as follows:

DC voltage: 1 V

Hide the form. Place this instance of *vdc* on the schematic, top-left to *nmos1v* instance.

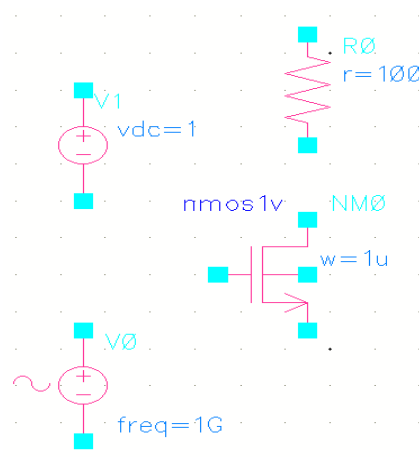
4. Press the **I** key.

Under the **analogLib** library, click in the cell text field and choose **res** in the Add Instance form change the parameter as follows:

Resistance: 100 Ohms

Hide the form. Place this instance of *res* on the schematic above the *nmo1v* instance.

5. You can use the keyboard arrow keys to move up and down the schematic. Your canvas should be similar to what you see in the following snapshot.



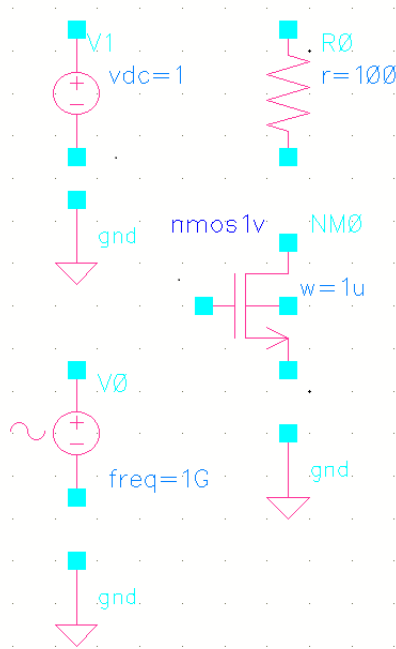
Adding Ground to the Schematic

1. Zoom out with the [key, so you have room to place the power and ground symbols.
2. Press **I** to display the Add Instance form
3. In the **Cell** text field of the Add Instance form, replace *vdc* with **gnd** under the same *analogLib* library.



4. Move the cursor into the schematic and place three *gnd* symbols below the source of *nmos1v*, below the negative terminal of *vsin* and below the negative terminal of *vdc*.
5. Then press the **Esc** key. This will *close* the Add Instance form.

Important: Pressing **Esc** cancels any active command in the schematic window.



Creating Pins to the Schematic

1. Use the **Create – Pin** menu command or the **Create Pin** icon to add pins to the design. You can also use the **P** key on the keyboard.

The Create Pin form is displayed.

2. **Type** the pin names in the Create Pin form in the order shown, leaving spaces between the pin names *vin* and *vout*.
3. Make sure the *Direction* field is set to **input** and the *Usage* field is set to the **schematic**.

4. Move your cursor into the *Design* schematic. You might have to click on the window banner to make it the active window.

Note: You can still move the pins around using the arrow keys while the pin placement feature is active.

Important: You can also rotate the pin 90 degrees each time you click the right mouse button.

- a. **Left-click** in the schematic where you want the *vin* pin to be.

Note: Notice that *vin* is no longer in the *Pin Names* field of the Create Pin form but is placed on the schematic.

- b. Similarly, **left-click** to place the *vout* input pin in another location. (After placing *vin*, you can directly place *vout* in another location by **left-click**.)

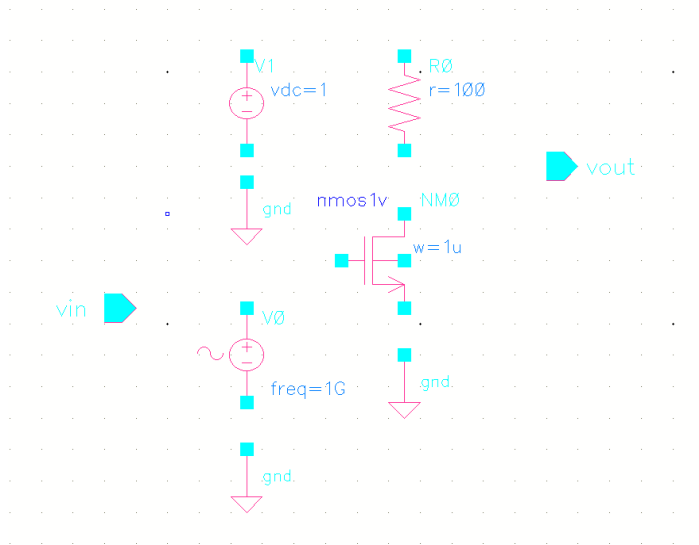
5. In the Create Pin form, change the *Direction* field to **output**.

Note: If you have hidden the form, press **F3** to display it again.

Move your cursor into the schematic window and **left-click** to place the *vout* pin in the proper location.

Note: If you do not change the direction to output before placing the pin, you might encounter errors later. If so, you can always edit the properties of the pin and change it to output.

6. Cancel the Create Pin form or press the <Esc> key to close or unselect the pin.
 - a. In the schematic window, choose **View – Zoom to Fit** or press **F**.
Move or rearrange the location of your components as needed. Your schematic window should be similar to what you see in the following snapshot.



Adding Wires to a Schematic

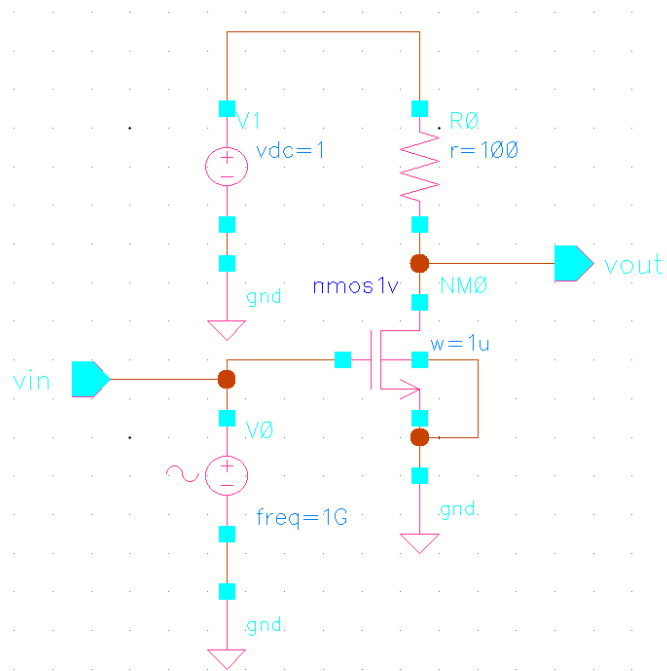
Add wires to connect components in the design.

1. Choose **Create – Wire (narrow)** from the pull-down menu. You can also use the **Create Narrow Wire** button from the Create toolbar section.
Note: Lowercase **w** is the bindkey for a *Narrow wire*, **Shift-W** for a *Wide wire*.
2. In the schematic window, click one of your components' terminals, such as the *drain* terminal of *nmos1v*, to make it the first point for your wiring.

Note: The wire router cannot always find a path between two points. If this happens, a dotted flight line is drawn between the points and establishes connectivity. If flight lines occur in your lab, delete them and click on the intermediate points to guide the router as your design is wired.

Follow the prompts at the bottom of the design window, move the mouse cursor, and **left-click** the destination point for your wire, such as n-mos gate. A net is routed.

- Continue wiring the schematic, as shown in the following snapshot.



The exact topology shown in the reference schematic is not as important as the connectivity.

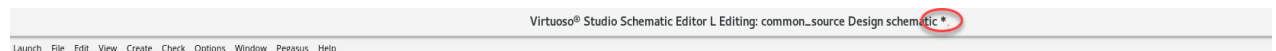
Caution: Be sure to connect the substrate for the `nmos1v` transistor to `gnd` respectively.

Important: If you make a mistake while wiring, select that section of the wire and press the <Delete> key to delete that part of the wire.

- Press **Esc** with your cursor in the schematic window to cancel the wiring when finished wiring.

Saving a Design

Note on the title bar that there is an asterisk next to the name schematic. This indicates that your schematic has been edited and needs to be saved.

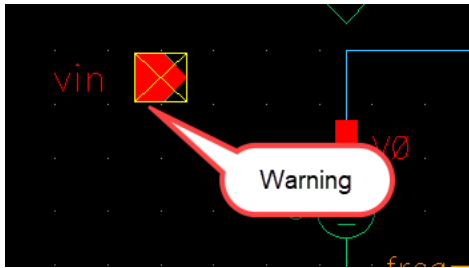


- Click the **Check and Save** icon in the schematic editor window.



- Or choose **File – Check and Save** or click **Shift + X**.

Note: A yellow box appears around all the unconnected pins and wires. If there are any, close the associated warning message and complete the connectivity.

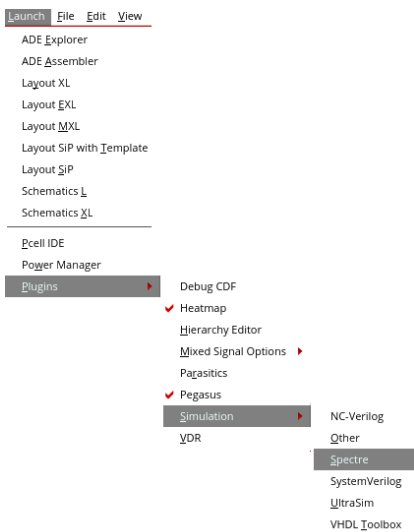


3. Check the CIW output area for any messages after a successful check and save.

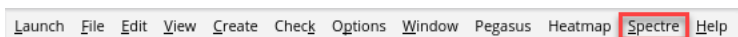
```
INFO (SCH-1170): Extracting "Design schematic"
INFO (SCH-1426): Schematic check completed with no errors.
INFO (SCH-1181): "common_source Design schematic" saved.
```

Netlisting the Design

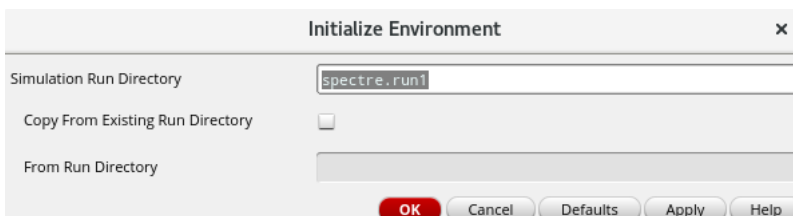
1. From the Schematic Editor window, choose **Launch – Plugins – Simulation – Spectre**.



Note: This attaches the Spectre Classic Simulator in the menu bar.



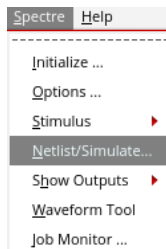
2. From the menu bar, choose **Spectre – Initialize**.



3. Click **OK** in the Initialize Environment form to create simulation result databases in the *spectre.run1* default simulation run directory.
 - a. Click **OK** in the next form that opens (shown in the following snapshot), verifying the design to be netlisted.



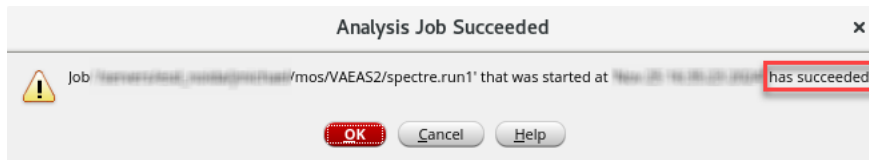
4. Choose **Spectre – Netlist/Simulate** to open the Netlist and Simulate form.



5. **Disable** the *Run Actions – simulate* option. Your form should look similar to the one you see in the following snapshot.



6. Click **OK**. In the *Analysis Job Succeeded* form that comes up, click **OK**.



7. View the netlist file in the *spectre.run1* directory by choosing **Spectre – Stimulus – Edit Netlist File**.



Try to associate the connectivity in the netlist along with your drawn schematic.

8. **Close** the netlist window, leave the *Design* schematic view and CIW session running, and proceed to the next lab.



Lab 1-2 Simulation Using ADE Explorer and Virtuoso VA

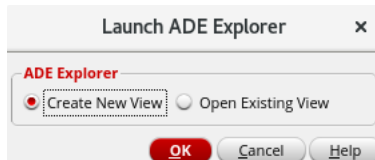
Objective: To simulate the current mirror circuit using ADE Explorer and Virtuoso VA.

In this lab, you

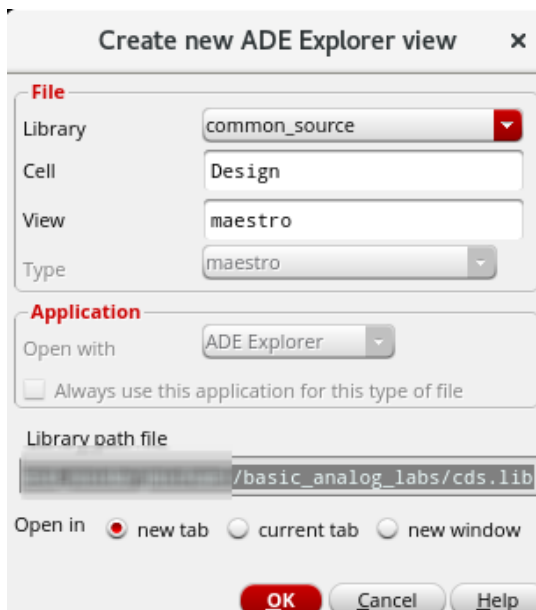
- ◆ Learn how to simulate the current mirror created in the previous lab by creating a new *maestro* view using Virtuoso ADE Explorer.

Starting the Virtuoso ADE Explorer and Creating a *maestro* View

1. Choose **Launch – ADE Explorer** from the *Design* schematic window to open the Launch ADE Explorer form.

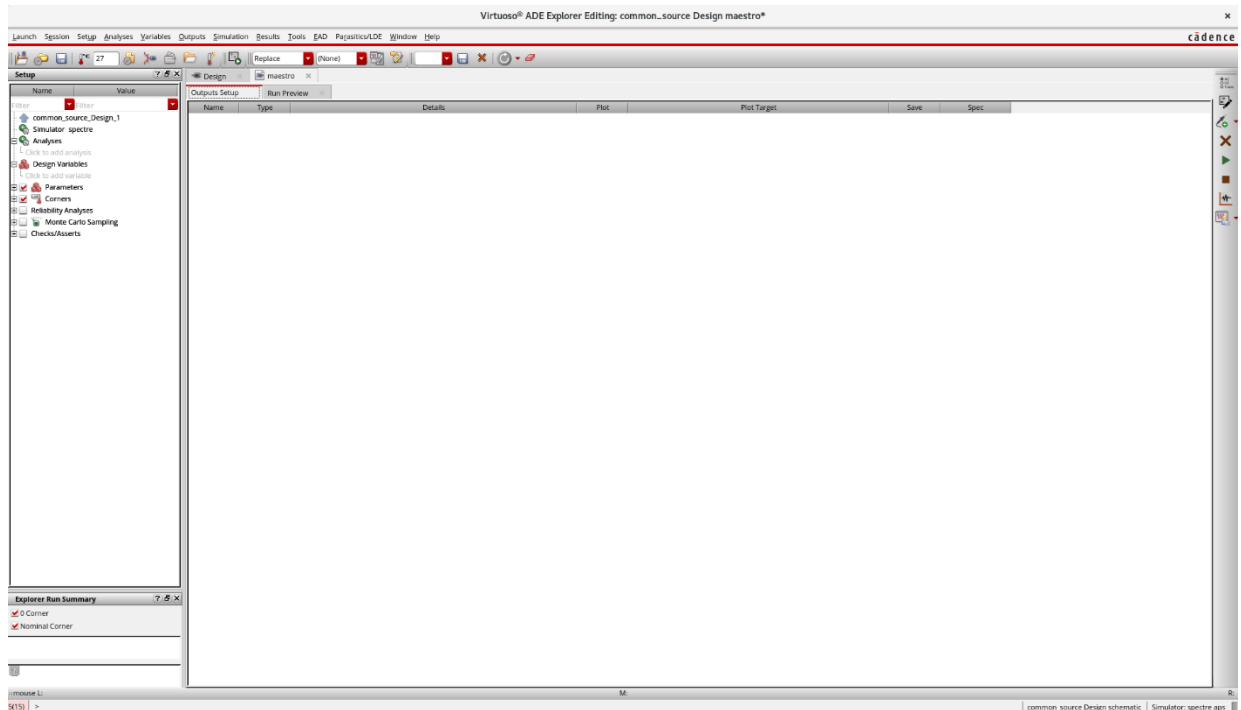


2. Select **Create New View** to open the **Create new ADE Explorer view** form.



3. The **maestro** view type is chosen by default, and so is the *Open with* the application, which is set to the **ADE Explorer**.

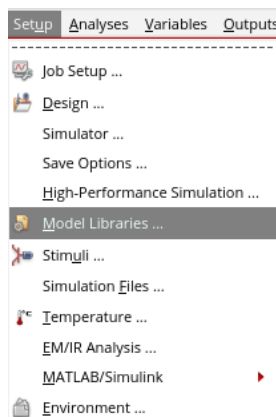
4. Leave the View name set to the default name, **maestro**. Choose to open in a new tab and click **OK** on this form to open the ADE Explorer environment in a new **maestro** tab.



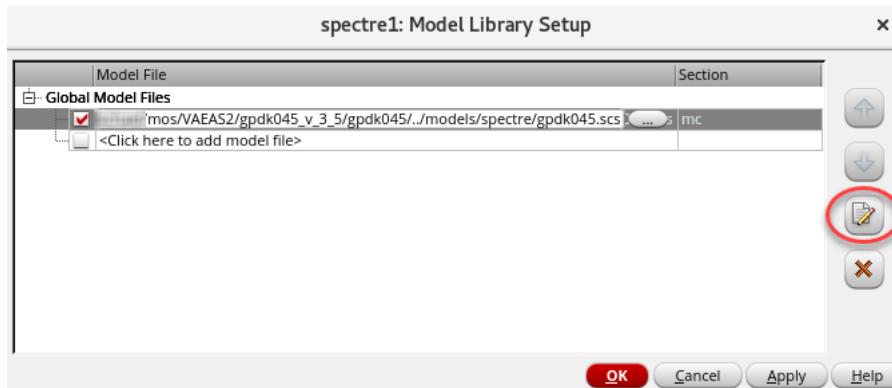
Setting the Model Libraries

The model library file contains the model files that describe the *nmos1v* device during simulation.

1. Choose **Setup – Model Libraries**.



The Model Library Setup form appears. You can see that there will be *gpd045 model library* added automatically. This is because, we have attached the **mirror** library to *gpd045* in the beginning.



- a. To view the model file, select the model file. Click the **Edit/View Selected Files** button.



Close the model file after viewing.

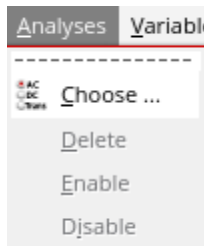
With the Model Library Setup form, you can include multiple models.

- b. On the Model Library Setup form, click **OK**.

Choosing Analyses

This part demonstrates how to select **tran analyses** to complete your circuit during the simulation. You can select and run dc analysis on the *current mirror* design.

1. In the Simulation window, choose **Analyses – Choose**



2. Or click the **Choose Analysis** icon.



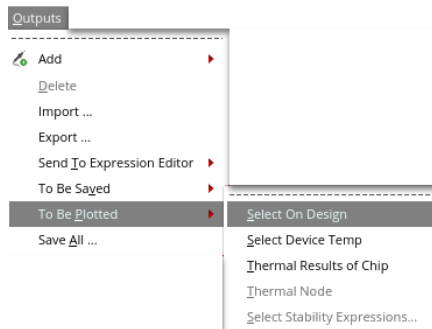
- Your form should look similar to the one you see in the following snapshot. Click on **OK**.

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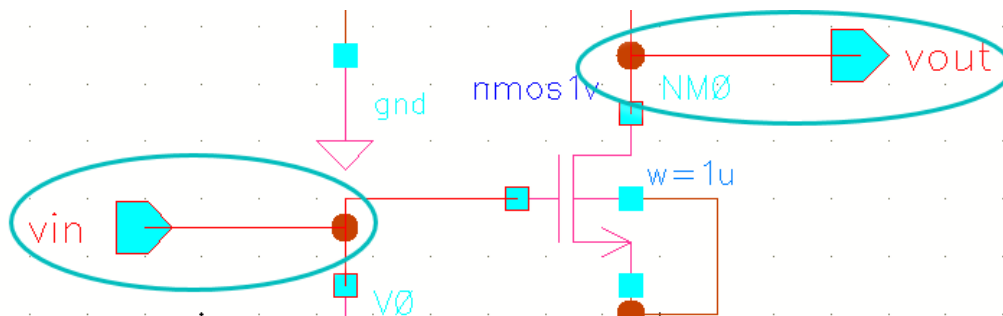
Saving and Plotting Simulation Outputs

The simulation environment is configured to save all public node voltages by default. It also saves the currents you select for plotting and saving.

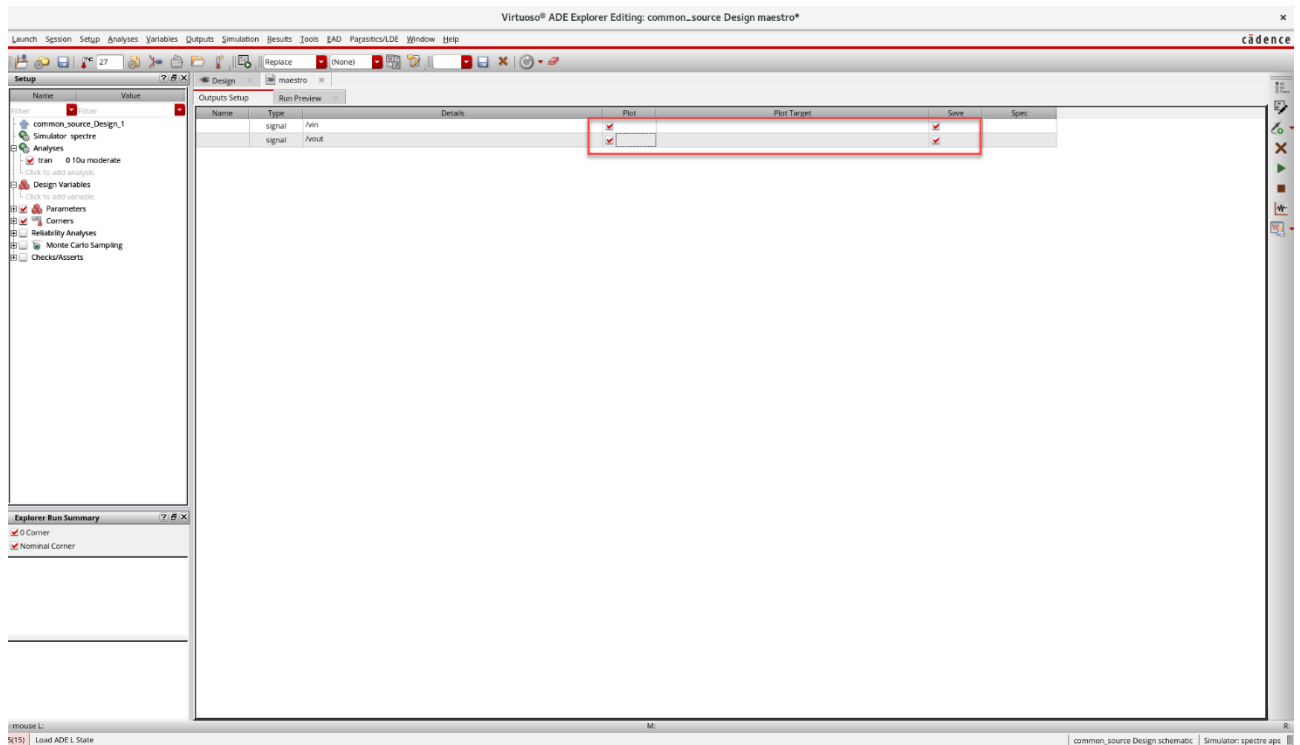
1. To select specific signals for saving and plotting, choose **Outputs – To Be Plotted – Select On Design** in the ADE Explorer window.



2. This brings you to the front of the schematic window for your selections.
 - a. Select the *net* coming from *vin* pin and the *net* connected to *vout* for plotting. After selections, the terminal gets highlighted.



3. Press **Esc** with your cursor in the schematic window to stop the selection process.
4. Move to the **maestro** tab. Enable the plot and verify whether it looks like the one in the following snapshot:

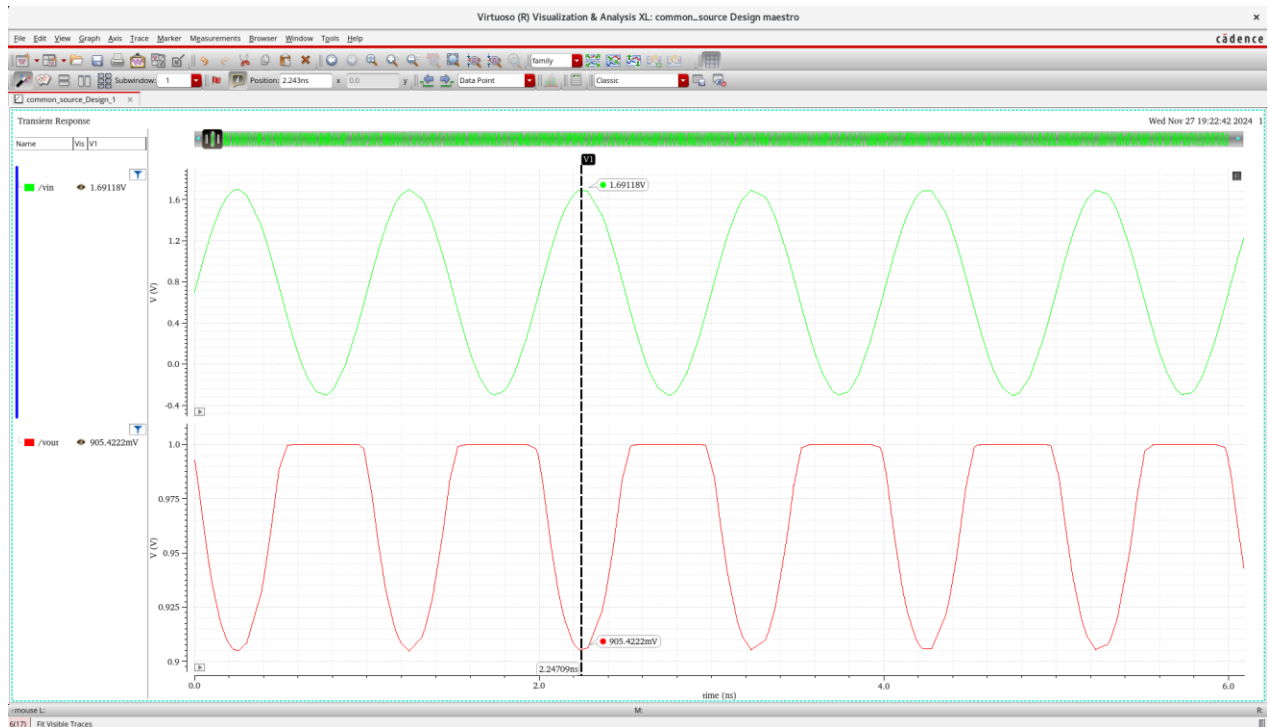


Running the Simulation

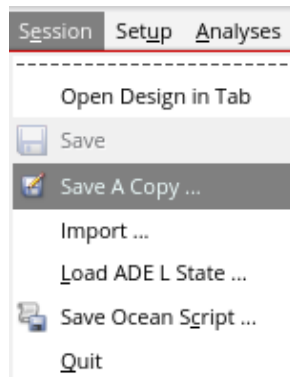
1. Choose **Simulation – Netlist and Run** or click the **Run Simulation** button  to recreate the netlist and run the simulation.
 - a. As an alternative to the previous step, you can choose **Simulation – Run** from the menu if you have already created your netlist.
 - b. A log file shows the progress of the simulation. This is called the *spectre.out* file and is stored in a hierarchical psf directory inside your *./simulation* directory.
 - c. The log file also lists warnings and errors if there are problems in the input data, which often is missing model files.

Tip: If you close this log file, you can open it again by going to the Simulation – Output Log.

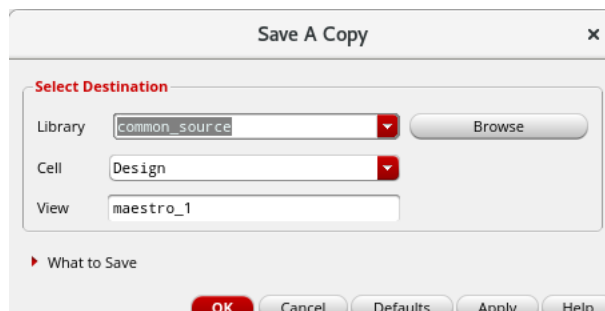
2. When the simulation finishes, the DC Response plot is plotted automatically in the Virtuoso Visualization & Analysis XL window.



- a. Click on the **Save** icon to save the graph. Move to Maestro window and save the simulation settings by clicking on the **Session – Save a Copy**.

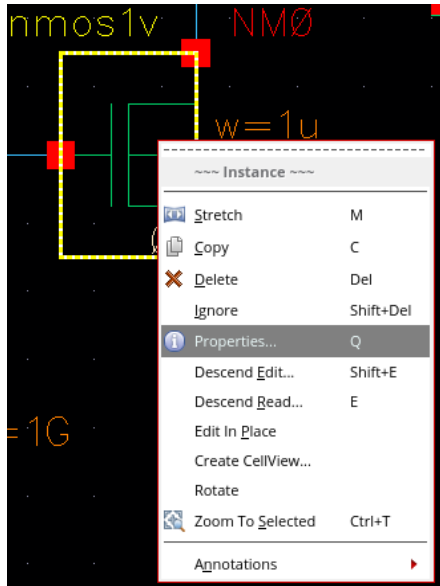


- b. In the **Save a Copy** form, name the view and save it.



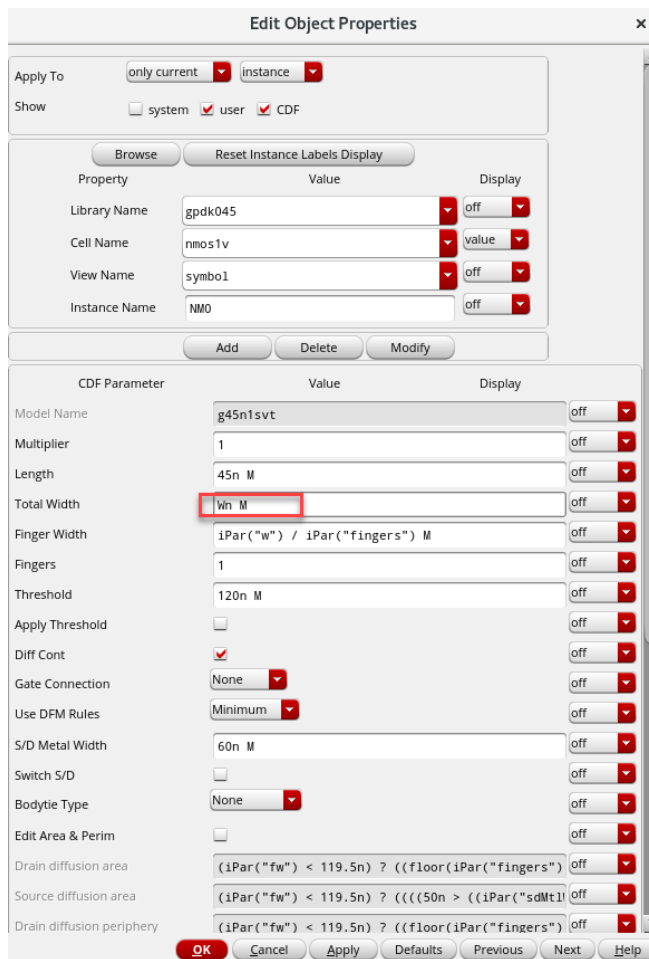
DC Parametric Analysis

1. For Parametric Analysis, move to the *Design* schematic view. Select the **nmos1v** instance and select **Properties** in the drop-down menu by **right-clicking** on the **nmos1v**.



2. On the **Edit Object Properties** form, type in the **Total Width** field as follows:
Total Width: Wn

- Your canvas should be similar to what you see in the following snapshot:



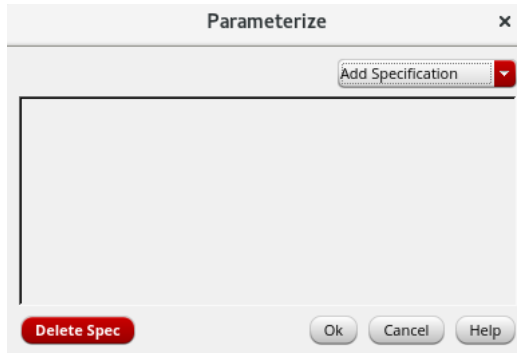
- Choose **File – Check and Save** or click **Shift + X**.
- Verify the CIW for warnings and errors.

Simulating the Parametric Analysis

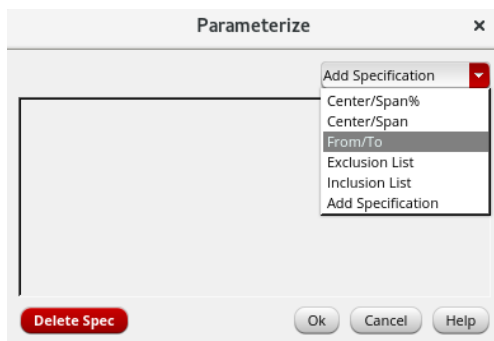
- Choose **Launch – ADE Explorer** from the *Design* schematic window to open the Launch ADE Explorer form.
- Move to the **Maestro** tab with the already filled settings of the previous analysis. Choose **Variables – Copy from Cellview** on the ADE Explorer menu. This includes the **Wn** variable in the **Design Variables** section.



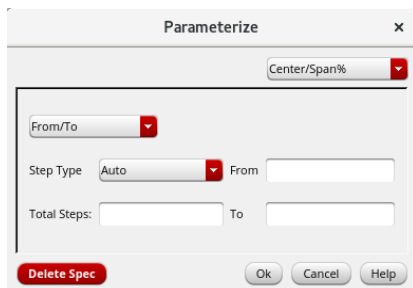
3. For the W_n , click on the value field to enable the button  of **Parameterize** form. Click on it to open the form once it is enabled.



4. In the drop-down of **Add Specification**, click on **From/To** to add the *Spec*.



5. In the **Step Type** drop-down menu, click on **LinearStepCount**.



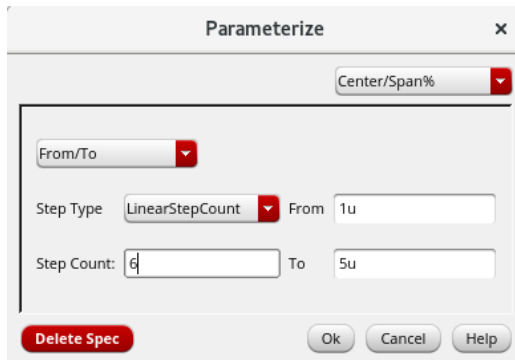
6. Type the values as follows:

From: 1u

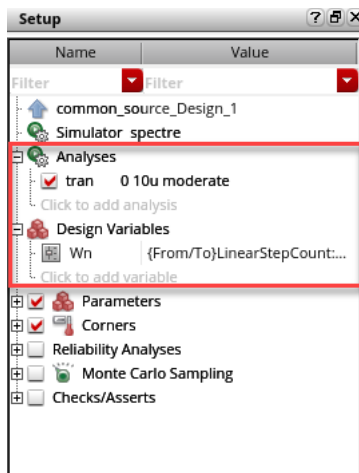
To: 5u

Step Size: 6

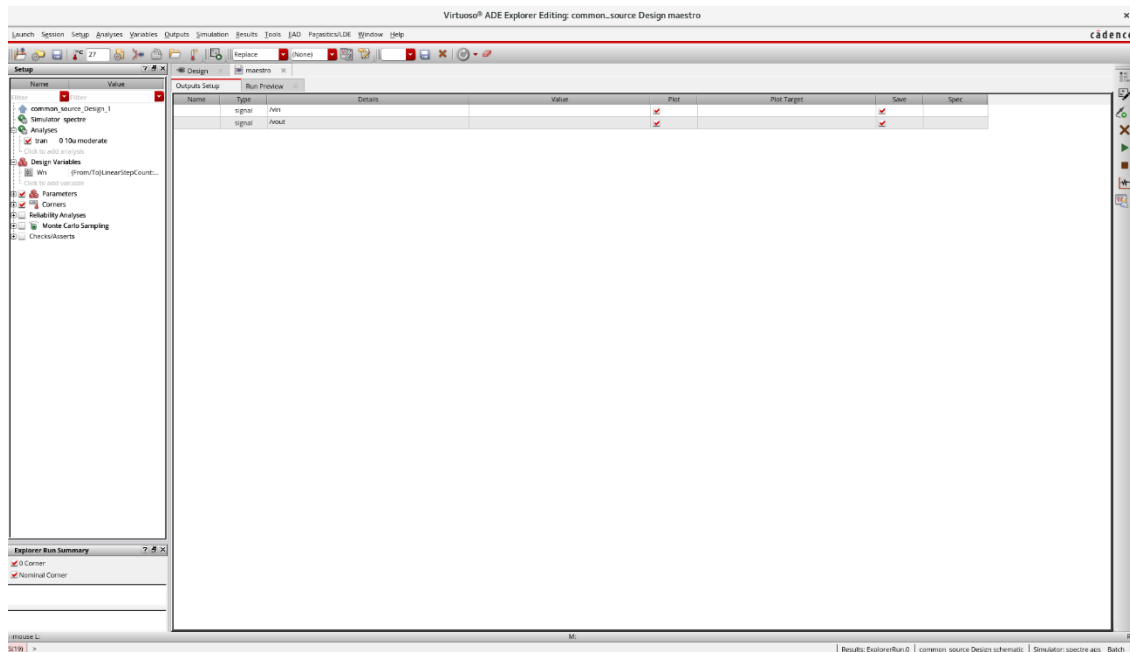
Your form should look similar to the one you see in the following snapshot. Click on **OK**.



- Keep the settings of the Analysis form as filled for the previous analysis. Your canvas should be similar to what you see in the following snapshot:



8. Keep the settings in the **Outputs Setup** in the **Maestro** tab, as filled for previous analysis. Your canvas should be similar to what you see in the following snapshot:



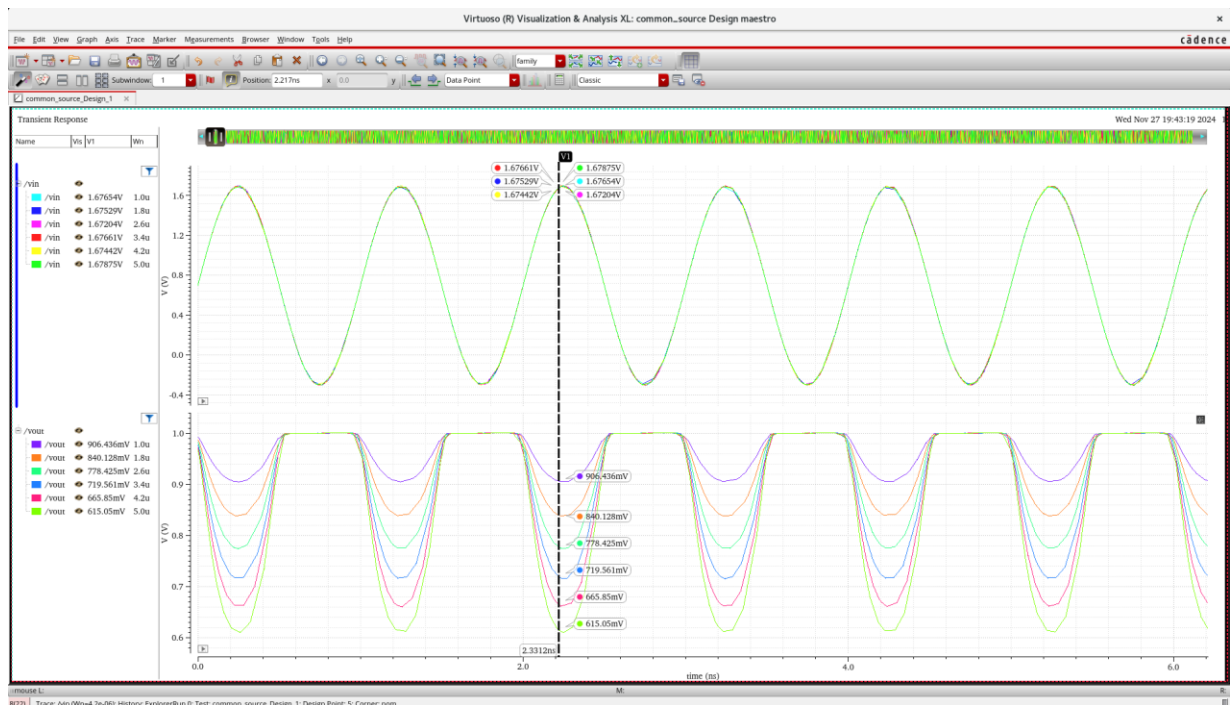
9. Choose **Simulation – Netlist and Run** or click the **Run Simulation** button  to recreate the netlist and run the simulation.
- As an alternative to the previous step, you can choose **Simulation – Run** from the menu if you have already created your netlist.
 - A log file shows the progress of the simulation for parametric analyses of W_p . This is called the *spectre.out* file and is stored in a hierarchical psf directory inside your *./simulation* directory.
 - The log file also lists warnings and errors if there are problems in the input data, which often is missing model files.

Tip: If you close this log file, you can open it again by going to the Simulation – Output Log.

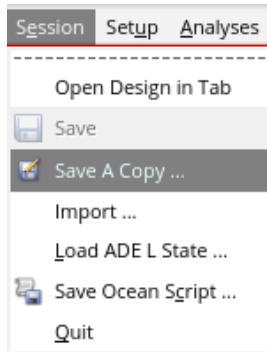
10. The Results tab of the Maestro shows the status of each trace or value of W_p according to the *Step Size* given in the **Parameterize** form.

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
1	common_sour...	/vin	pending			
1	common_sour...	/vout	pending			
2	common_sour...	/vin	pending			
2	common_sour...	/vout	pending			
3	common_sour...	/vin	pending			
3	common_sour...	/vout	pending			
4	common_sour...	/vin	pending			
4	common_sour...	/vout	pending			
5	common_sour...	/vin	pending			
5	common_sour...	/vout	pending			
6	common_sour...	/vin	pending			
6	common_sour...	/vout	pending			

11. When the simulation finishes, the DC Response plot is plotted automatically in the Virtuoso Visualization & Analysis XL window.



12. Click on the **Save** icon to save the graph. Move to Maestro window and save the simulation settings by clicking on the **Session – Save a Copy**.



13. In the **Save a Copy** form, name the view and save it.

